K-Means and K-Medoid Clustering

1. Introduction to Clustering

Clustering is one of the most important techniques in unsupervised learning, which aims to group a set of objects or data points into clusters such that data points within a cluster are more similar to each other than to those in other clusters. It is widely used in data mining, pattern recognition, image segmentation, market segmentation, and anomaly detection.

Clustering algorithms work without predefined labels or outputs. Instead, they use the inherent structure or similarity within the dataset to form groups. Among the different clustering techniques, K-Means and K-Medoid are the most popular partition-based algorithms.

2. K-Means Algorithm Overview

The K-Means algorithm is a centroid-based clustering method. It is one of the simplest and most efficient unsupervised learning algorithms used for partitioning a dataset into a set of *K* clusters, where *K* represents the number of clusters predefined by the user.

The goal of K-Means is to minimize the sum of squared distances between the data points and their respective cluster centroids. This approach helps identify patterns or structures in large datasets by finding groups of similar data points.

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Objective Function

The main objective of K-Means is to minimize the within-cluster sum of squares (WCSS):

$$J = \sum_{i=1}^{K} \sum_{x_j \in C_i} ||x_j - \mu_i||^2$$

Where:

K=number of clusters
Ci=cluster i
xj=data point in cluster i
μi=centroid (mean) of cluster i
||xj−μi||2=squared Euclidean distance between a data point and its cluster centroid

3. Centroid-Based Technique

The **centroid-based technique** refers to clustering algorithms that represent each cluster by a single central point — called the **centroid**. The centroid acts as a representative or prototype for the entire cluster.

In the context of K-Means:

- Each cluster is defined by its centroid.
- The centroid is the **mean of all points** belonging to that cluster.
- Each data point is assigned to the cluster whose centroid is closest to it (based on distance measures such as Euclidean distance).

The iterative process continues until the centroids stabilize and no further changes occur in the cluster assignments.

Distance Measure

The most commonly used distance metric in centroid-based methods is the Euclidean distance:

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + ... + (x_n - y_n)^2}$$

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Other metrics such as Manhattan or cosine similarity can also be used, depending on the dataset and context.

4. K Nonempty Subset Concept

In K-Means, the dataset is partitioned into **K nonempty subsets**. This means that:

- Each cluster must contain at least one data point.
- No cluster can be empty.

If any cluster becomes empty during the iterations, the algorithm reinitializes that cluster by assigning a random data point to it or by redistributing points from larger clusters. This ensures that all *K* clusters are active and meaningful.

5. Seed Point of Centroids of Cluster

The initialization of **seed points (centroids)** is a crucial step in K-Means. The initial centroids largely determine the quality of the final clusters and the speed of convergence.

Common Initialization Methods

1. **Random Initialization**:
    - Randomly select *K* data points from the dataset as initial centroids.
    - Simple but can lead to poor clustering results if the chosen points are not representative.
2. **K-Means++ Initialization**:
    - Improves upon random initialization by spreading out the initial centroids.
    - Procedure:
        - Select the first centroid randomly.
        - Choose subsequent centroids with a probability proportional to their squared distance from the nearest existing centroid.
        - Repeat until *K* centroids are chosen.
    - This reduces the chance of poor clustering due to bad initialization.

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3. **Forgy and Random Partition Methods**:
    - Forgy: randomly selects *K* observations as centroids.
    - Random partition: randomly assigns each observation to one of *K* clusters, then calculates centroids.

Good initialization helps avoid local minima and ensures faster convergence.

6. Partitioning in K-Means

K-Means belongs to the **partitioning family** of clustering algorithms. The key idea is to divide the dataset into *K* distinct, non-overlapping subsets or partitions, each representing one cluster.

Steps in Partitioning

1. **Initialization** – Choose *K* initial centroids.
2. **Assignment** – Assign each data point to the nearest centroid.
3. **Update** – Recalculate centroids as the mean of points in each cluster.
4. **Iteration** – Repeat assignment and update steps until centroids no longer change or a stopping criterion is met.

This process produces *K* partitions where:

- Each data point belongs to exactly one cluster.
- Clusters are non-overlapping.
- Every cluster is represented by its centroid.

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7. Steps of the K-Means Algorithm

The K-Means algorithm follows a straightforward iterative procedure:

Step 1: Initialization

Select *K* initial centroids (either randomly or using K-Means++).

Step 2: Cluster Assignment

Assign each data point to the nearest centroid using a distance measure (usually Euclidean distance).

For a data point xi and centroid μj:

Assign xi to cluster Cj if $|xi-\mu j||2$ is minimum

Step 3: Update Centroids

After all points have been assigned, compute the new centroid of each cluster as the mean of all data points belonging to that cluster:

$$\mu_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i$$

Step 4: Repeat

Repeat steps 2 and 3 until convergence, i.e., centroids stop changing significantly or the number of iterations reaches a threshold.

Step 5: Output

The final cluster assignments and centroids are returned.

8. Assigning Each Point to Cluster

This step involves allocating each data point to the cluster whose centroid is **closest** to it. This decision is based on the distance between the point and the centroids.

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Example

Consider three centroids:

μ1=(1,2)
μ2=(4,5)
μ3=(8,9)
For a data point x=(3,3) ,calculate:

$$d(x, \mu_1) = \sqrt{(3-1)^2 + (3-2)^2} = 2.23$$
$$d(x, \mu_2) = \sqrt{(3-4)^2 + (3-5)^2} = 2.23$$
$$d(x, \mu_3) = \sqrt{(3-8)^2 + (3-9)^2} = 7.81$$

The point x could be assigned to cluster 1 or 2 (tie), depending on implementation.

This assignment step ensures that each data point belongs to the cluster that minimizes its distance from the centroid.

9. Assigning Data Points to Nearest Point

This phrase refers to the **core logic** of K-Means — assigning every data point to the nearest cluster centroid. It's what makes the algorithm "centroid-based."

The "nearest point" here is the **centroid** of the cluster that has the minimum distance from the data point. This assignment is recalculated after each centroid update, ensuring clusters move toward optimal positions.

This iterative assignment minimizes the **intra-cluster variance** (distance between data points and their cluster centers) and maximizes the **inter-cluster variance** (distance between centroids).

10. Example of K-Means Clustering

Dataset Example

Suppose we have the following 6 two-dimensional points:

Point	X	Y
A	1	1
B	2	1
C	4	3
D	5	4
E	8	7
F	9	8

Let's apply K-Means with **K = 2**.

Step 1: Initialization

Choose centroids randomly:
Centroid 1 ($\mu 1$)=A (1,1)
Centroid 2 ($\mu 2$)= E (8,7)

Step 2: Assign Points

Calculate the distance of each point from both centroids and assign to the nearest one.
Cluster 1: A, B, C, D
Cluster 2: E, F

Step 3: Update Centroids

New centroid of Cluster 1 =mean of (1,1), (2,1), (4,3), (5,4) = (3, 2.25)
New centroid of Cluster 2=mean of (8,7), (9,8) = (8.5, 7.5)

Step 4: Repeat

Reassign and recompute until centroids no longer move.

The final clusters are:

- **Cluster 1**: A, B, C, D
- **Cluster 2**: E, F

The centroids converge, and clustering is complete

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11. Advantages of K-Means

1. **Simplicity** – Easy to understand and implement.
2. **Efficiency** – Works well for large datasets.
3. **Scalability** – Computationally efficient (O(nkt)).
4. **Convergence** – Usually converges quickly.
5. **Interpretability** – Centroids are intuitive and easy to visualize.

12. Limitations of K-Means

1. **Requires K as input** – Must specify the number of clusters in advance.
2. **Sensitive to initialization** – Poor initialization can lead to local minima.
3. **Assumes spherical clusters** – Not ideal for non-convex clusters.
4. **Sensitive to outliers** – Mean can be heavily influenced by extreme values.
5. **Different cluster sizes/densities** – K-Means struggles with irregular cluster shapes.

13. K-Medoids Algorithm

The **K-Medoids algorithm** is a variant of K-Means that aims to overcome its sensitivity to outliers. Instead of using centroids (which are means of points), K-Medoids uses **medoids**, which are **actual data points** that are most centrally located within each cluster.

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Definition

A **medoid** is the object within a cluster whose average dissimilarity (distance) to all the other objects in the cluster is minimal.

Key Difference from K-Means

Feature	K-Means	K-Medoids
Cluster Center	Mean (not necessarily a real point)	Actual data point
Sensitivity to Outliers	High	Low
Objective Function	Minimizes squared distances	Minimizes pairwise dissimilarities
Robustness	Less robust	More robust

14. Steps of K-Medoid Algorithm

1. **Select K data points randomly as medoids.**
2. **Assign each data point** to the cluster with the nearest medoid.
3. **Calculate total cost** (sum of distances from points to medoid).
4. **Swap medoid** with a non-medoid point and compute new cost.
5. **If total cost decreases**, keep the new medoid; else revert.
6. Repeat until no changes occur in medoid positions.

15. Comparison between K-Means and K-Medoids

Aspect	K-Means	K-Medoids
Cluster Representation	Mean of points	Actual data point
Robustness to Noise	Low	High
Computation	Faster	Slower (more pairwise comparisons)
Used For	Large datasets	Small to medium datasets
Distance Metric	Typically Euclidean	Any dissimilarity measure

16. Applications of K-Means and K-Medoid

1. **Market Segmentation** – Grouping customers based on behavior.
2. **Image Segmentation** – Identifying distinct regions in images.
3. **Document Clustering** – Organizing text or articles into categories.
4. **Anomaly Detection** – Finding unusual patterns.
5. **Genomic Data Analysis** – Grouping genes with similar expression.

17. Practical Considerations

Choosing the Value of K

Selecting the right number of clusters is crucial. Common techniques:

- **Elbow Method** – Plot WCSS vs. K and look for an “elbow” point.
- **Silhouette Score** – Measures how well each point fits into its cluster.
- **Gap Statistic** – Compares total intra-cluster variation with that expected under a null reference distribution.

18. Conclusion

The **K-Means algorithm** is a fundamental unsupervised learning technique used to partition data into *K* clusters by minimizing intra-cluster variance. Its simplicity, speed, and interpretability make it highly popular across domains. The **K-Medoid algorithm** improves robustness by using actual data points as cluster centers, making it better for datasets with outliers or non-Euclidean distance measures.

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Together, these algorithms form the foundation for many advanced clustering methods used in data science and machine learning today.